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Fire Fighting Robot

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ABSTRACT

Fire incident is a disaster that can potentially cause the loss of life, property damage and permanent disability to the affected victim. They can also suffer from prolonged psychological and trauma. Fire fighters are primarily tasked to handle fire incidents, but they are often exposed to higher risks when extinguishing fire, especially in hazardous environments such as in nuclear power plant, petroleum refineries and gas tanks. They are also faced with other difficulties, particularly if fire occurs in narrow and restricted places, as it is necessary to explore the ruins of buildings and obstacles to extinguish the fire and save the victim. With high barriers and risks in fire extinguishment operations, technological innovations can be utilized to assist firefighting. Therefore, this paper presents the development of a firefighting robot dubbed QRob that can extinguish fire without the need for fire fighters to be exposed to unnecessary danger. QRob is designed to be compact in size than other conventional fire-fighting robot in order to ease small location entry for deeper reach of extinguishing fire in narrow space. QRob is also equipped with an ultrasonic sensor to avoid it from hitting any obstacle and surrounding objects, while a flame sensor is attached for fire detection. This resulted in QRob demonstrating capabilities of identifying fire locations automatically and ability to extinguish fire remotely at particular distance. QRob is programmed to find the fire location and stop at maximum distance of 40 cm from the fire. A human operator can monitor the robot by using camera which connects to a smartphone or remote devices.

INTRODUCTION

A robot is an automated device which performs functions usually attributed to humans or machines tasked with repetitive or flexible set of actions. Numerous studies have shown that robot can be beneficial in medicine [1], rehabilitation [2-6], rescue operation [7, 8] and industry [9]. Over the years, robotics has been introduced in various industries. The industrial robots are multi-function manipulators designed for more specialized materials, divisions, gadgets or devices through various programmatic movements to perform various tasks [10]. In line with the Fourth Industrial Revolution (4IR), there is demand for a one system that can control, communicate and integrate different robots regardless of their types and specifications.

Machine learning has also heated up interest in robotics, although only a portion of recent development in robotics can be associated with machine learning. Recent robotic development project has embedded machine learning algorithms [11-15] to increase the intelligence in robots. This will increase the productivity in industry while reducing the cost and electronic waste in a long run.

Studies on the use of humanoid robots are actively carried out to minimize firefighters' injuries and deaths as well as increasing productivity, safety, efficiency and quality of the task given [16]. Robot can be divided into several groups such as Tele-robots, Telepresence robots, Mobile robots, Autonomous robots and Androids robots. Telepresence robot are similar to a tele-robot with the main difference of providing feedback from video, sound and other data.

Hence, tele-presence robots are widely used in many fields requiring monitoring capability, such as in child nursery and education, and on improving older adult's social and daily activities [17, 18]. Mobile robot is designed to navigate and carry out tasks with the intervention of human beings [19, 20]. Meanwhile, autonomous robots can perform the task independently and receive the power from the environment, as opposed to android robots which are built to mimic humans [21].

In this paper, a firefighting robot is proposed. The main function of this robot is to become an unmanned support vehicle, developed to search and extinguish fire. There are several existing types of vehicles for firefighting at home and extinguish forest fires [22]. Our proposed robot is designed to be able to work on its own or be controlled remotely. By using such robots, fire identification and rescue activities can be done with higher security without placing fire fighters at high risk and dangerous conditions. In other words, robots can reduce the need for fire fighters to get into dangerous situations. Additionally, having a compact size and automatic control also allows the robot to be used when fire occurs in small and narrow spaces with hazardous environments such as tunnels or nuclear power plants [23, 24].

Thermite and FireRob are two current available fire fighter robots that have been used widely in industry. Thermite (produced by Howe and Howe Technologies Inc) is a firefighting robot that uses a remote control and can operate as far as 400 m. It can deliver up to 1200 gpm of water or 150 psi of foam. The size of this robot is 187.96 cm x 88.9 cm x 139.7 cm. This robot powers up to 25 bhp (18.64 kW) using a diesel engine. The main component in the design of this robot are multi-directional nozzle that is backed by a pump that can deliver 600 gpm (2271.25 l/min). This robot is designed for use in extreme danger areas, such as planes fires, processing factories, chemical plants or nuclear reactors [25].

Challenges in Firefighting

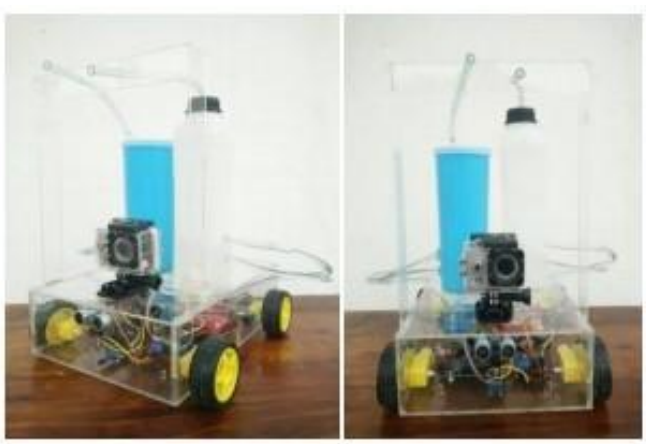
Firefighting is one of the most dangerous professions due to its unsafe and constantly changing environment. Firefighters may be required to operate under conditions with a high level of uncertainty and must make time-critical decisions using insufficient information [5]. Despite thorough preparation, firefighters continue to face challenges while operating in structural fires. These challenges are the result of uncertainties of structural integrity, unpredictability of fatal events such as flashovers and backdrafts, and getting disoriented or lost when entering buildings with unknown layouts.

METHODOLOGY

The methodology is divided into three parts. The first part is on the mechanicals schematics, followed by hardware description and the finally on the programming design. All parts were assembled together and experiments were then performed to determine the optimal distance of QRob to extinguish the fire were carried out. A. Mechanical Design Structure Google SketchUp software and AutoCad were used to produce 3D and 2D schematic diagram. For the main structure of the robot, to get the preferred movement and speed, QRob have two wheels at rear side and two wheels at front side.

The wheels have the ability to stabilize the robot and make rotation until 360 degrees. The body of the robot is made from acrylic plate to protect the electronic circuit. The acrylic sheet is resistant to heat of up to 200 º C. Thisgives the ability to use and work with (cut and drill). The body of acrylic chassis contains holes that make it easier to mounting of various type of sensors and other mechanical components.

The ultrasonic sensor and flame sensor were installed at front of the robot to avoid hitting any obstacles and to detect the fire respectively. In addition, mini camera was installed in front side of the robot to monitor the way and condition of the location and is linked to the smart phone. The structure of fire distinguisher robot is shown in Fig. 1 and Fig. 2.



HARDWARE IMPLEMENTATION

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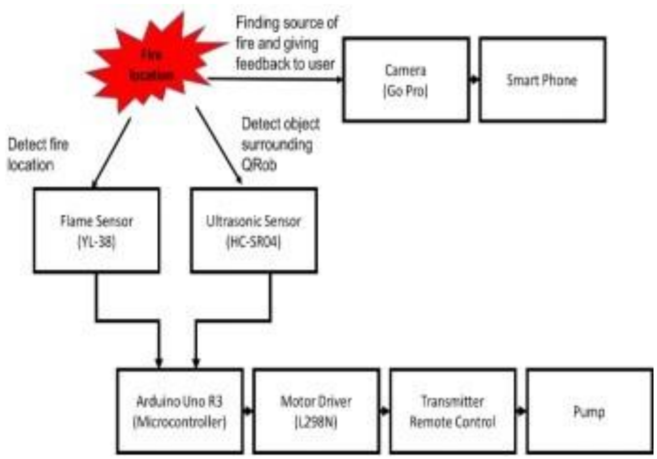
The electronic part is one of the vital parts in the development of QRob. It includes the several types of sensors, microcontroller, DC motor with wheel, Transmitter and Remote control and Water pump. Fig. 3 shows the block diagram of the QRob operation which consists of flame sensor and ultrasonic sensor as input of the system. Arduino Uno is used as a microcontroller that connected with other components. Motor Driver (L298N) is used to activate the moving of the gear motor while Transmitter Remote Control will give output of the system. Flowof water and fire extinguisher were pump after being controlled by the operator. On the other hand, the operator can monitor the robot movements by using camera (Go Pro) which connects to a smartphone.

Flame sensor: In most firefighting robots, fire sensors perform an essential partin investigations, which are always used as robot eyes to discover sources of fire [1]. It can be utilized to identify fire based on wavelength of the light at 760 nm to 1100 nm. The detection angle and distance are roughly 60 degrees and distance 20 cm (4.8V) to 100 cm (1V) respectively. Flame sensor has two signal pins that are Digital Output (DO) and AnalogOutput (AO). DO pins will give two kind of information that it's has flame or non-flame while AO pins will detect exact wavelength of different light.



Ultrasonic sensor:

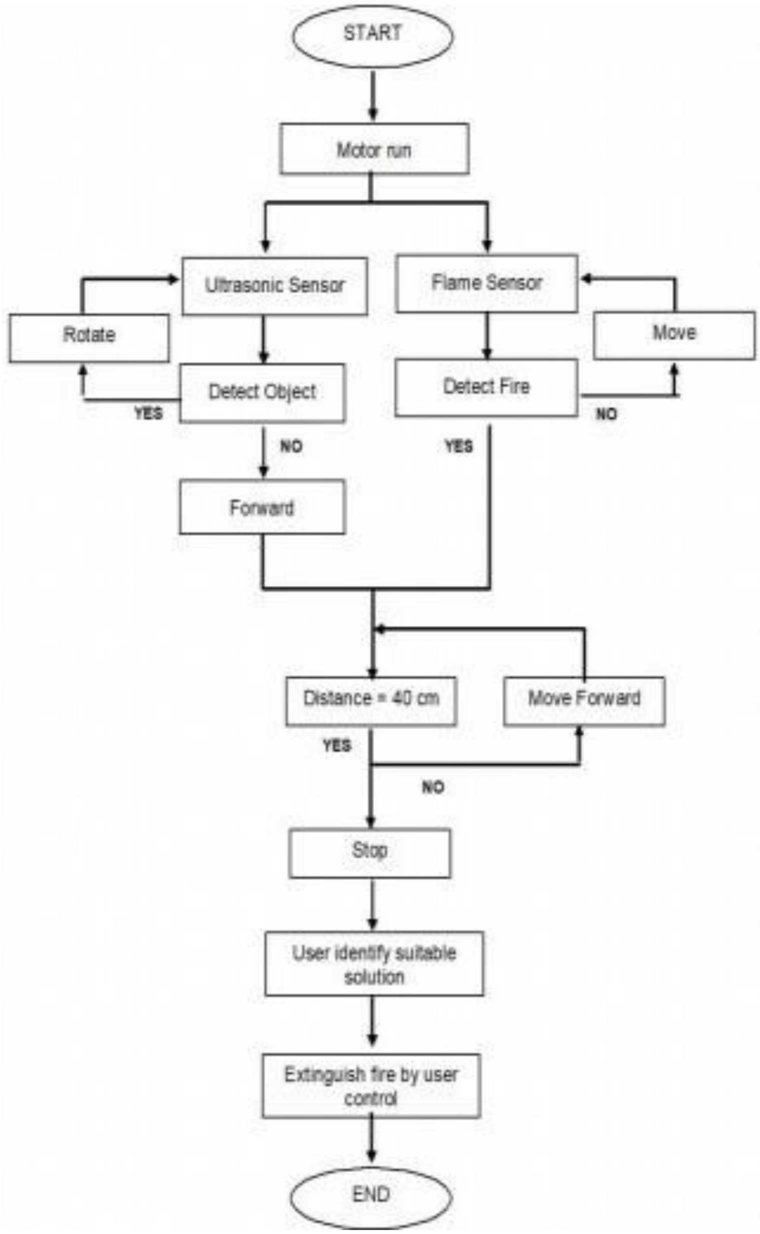
One of the most crucial aspect in inventing an autonomous target detection robot is a barrier and obstacle avoidance. A sensor must be compact, low cost, simple to produce and functional on a larger scale. Moreover, it should be able to sense things with enough limits to let robots to react and travel appropriately. The existing sensors that suit all these requirements are ultrasonic sensors. The HCSR04 ultrasonic sensor is utilized in this study to determine the distance within the range of 2 cm to 400 cm with an angle 15 degrees.

This sensor transmits waves into the air and receive reflected waves from the object. It has four output pin such as reference voltage (VCC) (operate around 5V), ground pin (GND), digital output (DO) and analog output (AO). 3) DC motor with wheel: DC geared motor with rubber wheel are suitable material for this project. This DC motor are suitable to replace 2 WD and 4 WD car chassis. The working voltage for DC motor is around 5V to 10 V DC. While the ratio of the gear is 48:1. Suitable current for this motor is 73.2 mA. DC motor is used to move the robot to the fire location.

Water Pump:

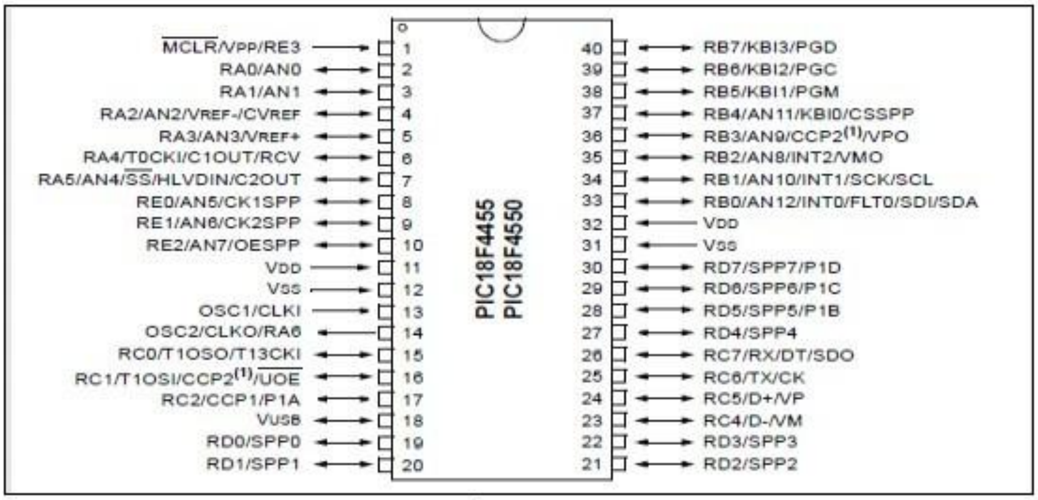
The water pump is important part in this robot as it will pump water or soap to extinguish the fire depending on the class of fire that occurs. Small-size and light-weight category of water pump has been selected for use in this project. Moreover, it has low noise, high effectiveness and minimal power consumption. The optimal voltage for this water pump is 6V. Working voltage for this water pump is around 4V to 12V with the working current 0.8A.

Transmitter and remote control: In this study, the wireless remote control transmitter and receiver with 4 control modes will be used. Model number of this receiver or remote is S4C-AC110. This remote have four buttons. Theoperating voltage for this remote control is AC 100 – 120 V, while the working voltage range of relay are AC 110 – 240 V or DC 0 – 28 V. The model numberof the transmitter is C-4. The distance of the remote control is 100 m or 300ft. Power supply for this transmitter are 12 V. The transmitting frequency is 315 MHz / 433 MHz. By utilizing the transmitter and remote control, QRob can be controlled from distant places where the operator who controls it will be in a safe place while the robot will enter into a dangerous fire area.



PIC18F4550 Microcontroller

A microcontroller is an inexpensive single chip computer, the entire computer system lies within the confines of the integrated circuit chip. Low cost, lowpower consumption, easy handling and flexibility make PIC18F4550 applicable in areas where microcontrollers had not previously considered. It has 33 I/O pins. Figure 3.6 shows the block diagram of PIC18F4550.



The description of the PIC18F4550 Ideal for low power (nanoWatt) and connectivity applications that benefit from the availability of three serial port FSUSB(12Mbit/s), I²C™ and SPI™ (up to 10 Mbit/s) and an asynchronous (LIN capable) serial port (EUSART). Large amounts of RAM memory for buffering and Enhanced Flash program memory make it ideal for embedded control and monitoring applications that require periodic connection with a (legacy free) personal computer via USB for data upload/download and/or firmware updates. Table 3.1 show to parameter of the PIC18F4550.

Table 3.2 : Parameter of PIC18F4550

PARAMETER	
Pin Count	40
Program Memory	32 Flash (KB)
CPU Speed (MIPS)	12 RAM Bytes 2,048
Data EEPROM	256 bytes
Digital Communication Peripherals	1-A/E/USART, 1-MSSP(SPI/I2C)
Capture/Compare/PWM Peripherals	1 CCP, 1 ECCP
Timers	1 x 8-bit, 3 x 16-bit
ADC	13 ch, 10-bit
Comparators	2
USB (ch, speed, compliance)	1, Full Speed, USB 2.0
Operating Voltage Range	2 to 5.5 (V)

Servo Motor

RC servos are hobbyist remote control devices servos typically employed in radio-controlled models, where they are used to provide actuation for various mechanical systems such as the steering of a car, the flaps on a plane, or the rudder of a boat.

Problem Statement

Fire environments are dangerous and constantly changing. Firefighters can become disoriented or lost when engaging in interior attacks on structural fires in compartmentalized buildings which can lead to risk of injury or death. Robots can be used to provide real-time data and a map of a building's complex, unpredictable layout. The goal of this project is to design and build a robot that will provide firefighters with additional information about a fire environment to help them make more informed decisions when fighting a fire. The robot must be compact and quick to deploy with a heat, water, and impact resistant chassis to function in unpredictable firegrounds. The remote-controlled robot will return a real-time video feed and a heat map from inside the building.

Contributions

In this project, we created a Firefighting Remote Exploration Device (FRED)to assist firefighters by gaining more information about fireground conditions. The team researched fireground hazards and designed a compact remote-controlled robot that can be deployed in a fireground to sense temperature and wirelessly transmit thermal data to a member of the firefighting crew. The robot is robust and human-portable for ease of deployment and is designed with heat- resistant, water-resistant, and impact-resistant features to face the hazards of a fireground. Material decisions were made with the goal of keeping the robot inexpensive in order to be affordable for fire departments. The robot also does not require a tether, to avoid getting in the way of firefighters. FRED is equipped with a camera, thermal arrays, and distance sensors; it is able to gather and send real-time thermal information and live video feed of the environment back to its operator. FRED also has a unique drivetrain specifically for unpredictable fireground terrain; it is driven by two passive transformable whegs that provide the smooth drive of standard wheels but the obstacle scaling capabilities of whegs.

Basement Fires

Basement fires are not only considered to be one of the most challenging operations to encounter, but they are also known to result in property loss and fatalities. Firstly, they are difficult to identify because the fire may appear to be coming from the ground floor . Secondly, 4 basements are difficult to access as smoke and heat builds up immense pressure that forces firefighters away. Because many basements have no walls or doors and the contents of a basement are often highly flammable, fires can develop rapidly and egress routes are limited. Lastly, basement fires can be very dangerous as basement ceilings are susceptible to collapse during a fire bringing risks to both firefighters on the ground floor as well as the basement [11].

High-rise Building (HRB) Fires

High-rise building (HRB) fires pose two types of challenges for firefighters both in and outside the building. The main challenge presented to firefighters is the height of the building. For active firefighting outside the building, equipment does not always reach the top floors. There is an upper limit on the amount of water pressure that can be applied by fire engine hoses which means water cannot reach the upper floors of HRBs. Additionally, firefighters inside HRBs have to carry heavy equipment up many flights of stairs while simultaneously evacuating any remaining occupants [12].

Once a firefighting unit is at the source of the fire, the second challenge is to fight it. Fires in high-rise hallways are problematic because they are defined as enclosed spaces (i.e., they are long and narrow, and they lack windows and open doors). According to Mora’s 2005 study, survivors of high-rise fires have described flashovers as “blowtorch-like” due to the winds blowing into the fire from an open hallway door [10]

Disaster Robotics

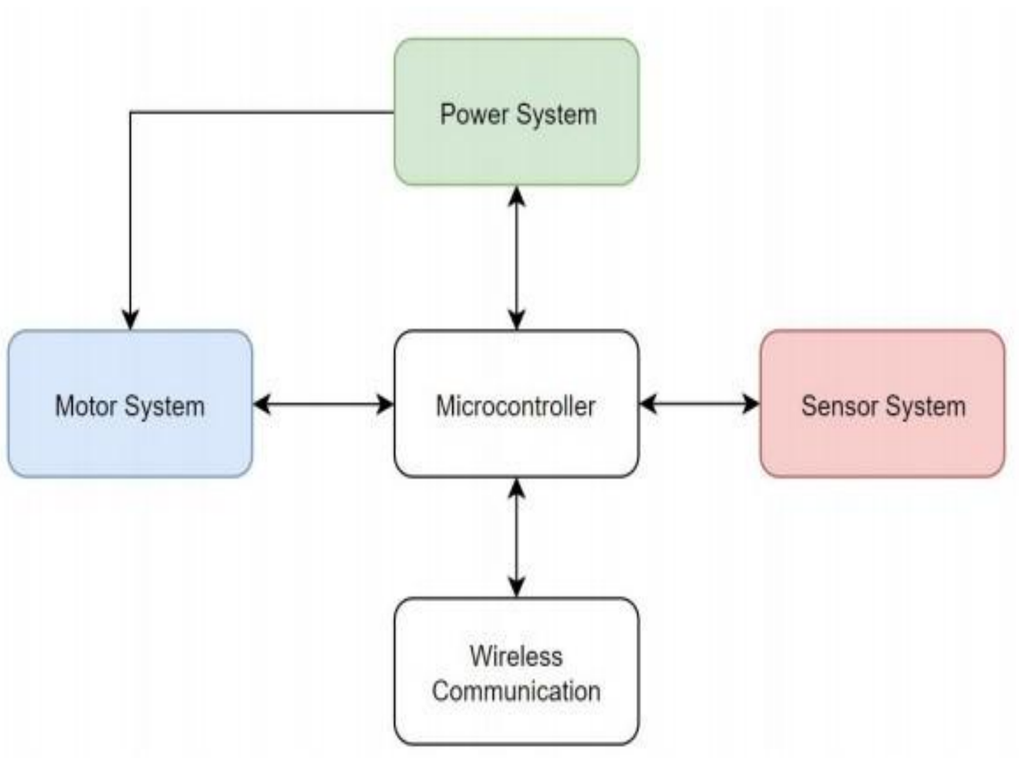
Currently, many robotic solutions exist that can aid workers in public safety jobs. Robots are machines that are commonly used to replace humans in labor- intensive, repetitive, or dangerous tasks [21]. Without prior information on the building layout during a structural fire and without real-time information on the current state of firefighters and the environment, firefighters put their lives atrisk when entering a fireground. While robotics can mitigate the risks posed to firefighters and the victims of a fire, most robotic solutions are large and heavy which can make it difficult to navigate a fireground. The following disaster relief robots have been developed to assist in the fireground.

Hoya Firefighting Robot

In 2009, the South Korean Hoya Robot Company designed a compact firefighting spy robot (Figure 6) that could be thrown and withstand a fall impact of 15 meters high, as well as withstand temperatures up to 160ºC for a 30-minute duration. The robot can be easily held by hand as the chassis has a height of 12.5 centimeters (cm) and weighs approximately two kilograms (kg). By using sensors to monitor the environment (measuring temperature, oxygen, and carbon dioxide concentrations) and having the ability to recognize voices, this remote-controlled device is an efficient means of exploring an unknown layout and providing significant information to firefighters to plan their actions accordingly [22][23]. Additionally, both the tmdwl [24] and hoyarobot [25]YouTube channels have videos to showcase the robot’s functionality and demonstrate the device operating side-by-side with actual firefighters [26].

System Design

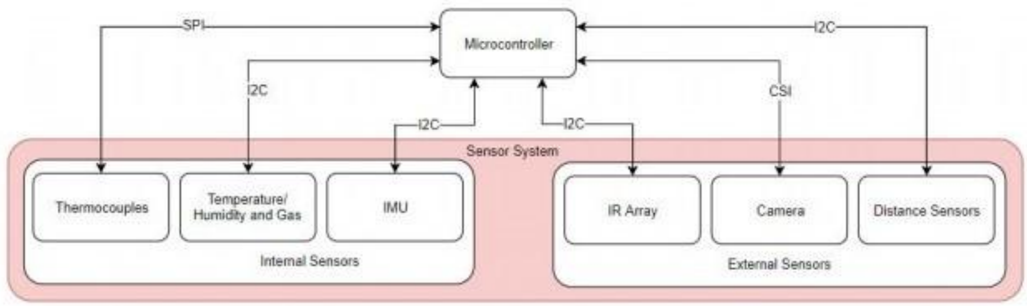
This section contains the complete electrical system design of the firefighting robot. Prior to the implementation stage, it is vital that the correct choices are made to ensure optimal functionality of the device. Using the requirement specifications and the background research, a high-level block diagram was designed (Figure .



As shown in the figure above, the high-level block diagram is composed of five different modules: a microcontroller unit (MCU), a power system, a motor system, a sensor system, and wireless communication. In the following sections, the design choices of each module will be thoroughly described using pre-made assumptions, comparison charts, and scientific reasoning.

Sensor System

As specified previously, the robot must: 1. survive a high temperature environment of 300°F for 15 minutes; 2. detect the temperature and heat flux of its environment; 3. and provide a 2D heat map overlay of the environment it is traversing. Given that the robot will be teleoperated and a map of the room will be provided, the sensor system will be composed of two different subsystems – an external sensor system that is used to detect information from the environment, and an internal sensor system that is used to detect changes in the robot.



External Sensor System

The external sensor system is made up of three main types of sensors: a camera, distance sensors, and infrared (IR) arrays. Figure 29 is a depiction of the external component configuration on the robot. Since the robot will be remote controlled from outside the building, it is necessary that the driver has a means to visually perceive the environment. Therefore, a camera on the front of the robot will be utilized. Although the robot is teleoperated and a map of the field is provided, distance sensors are necessary for localization. These will be placed on the front and sides of the robot to help understand where the robot is located relative to its position on the map. The IR array sensors will be used to detect both room temperatures and hotspots, and they will be placed around the robot. The front IR array will mainly be utilized to aid the driver in identifying areas by temperature.

GENERAL REQUIREMENTS

The device is described as autonomous mean of firefighting in houses and any civil buildings. Fire Protection Robot [11] (USA) – another competition project, developed for «15th Annual Trinity College Fire Fighting Robot Competition». Robot has more complex organization, than one, shown above and is orientedfor solving larger variety of tasks. The main system’s advantages are:
more complex algorithms, used for fire detection.

- using of sound sensor for activating.
- presence of some additional navigation sensors.
- The main disadvantages are: low-efficiency computer;
- low-power chassis;
- absence of home-return algorithm;
- absence of mapping;
- Firefighting Robot [12] is an American

Trinity College development, that was only on early-prototype stage (in 2008). It was supposed to this robot to be an autonomous device, with 15 minutes limited working time, after which it will return to the supply station. This approach is one of the best variants for firefighting in houses and non-industrial buildings. The main disadvantages are:

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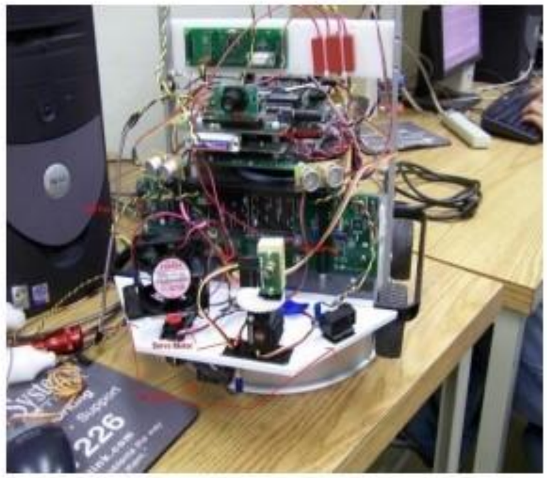
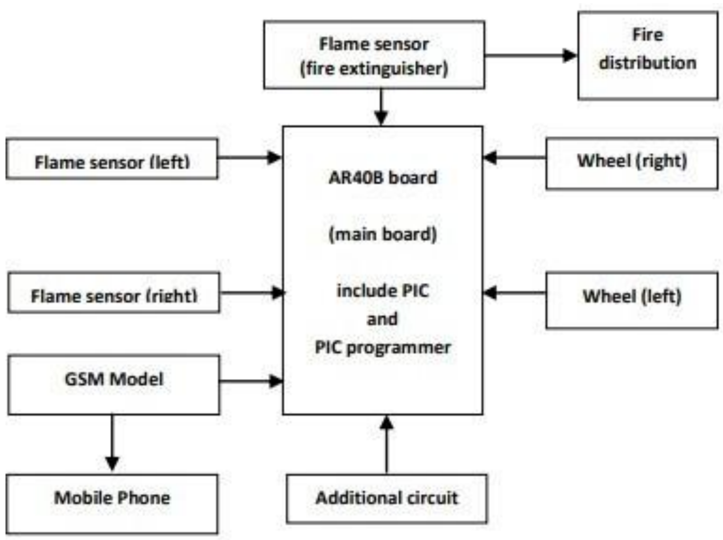


Fire Protection Robot

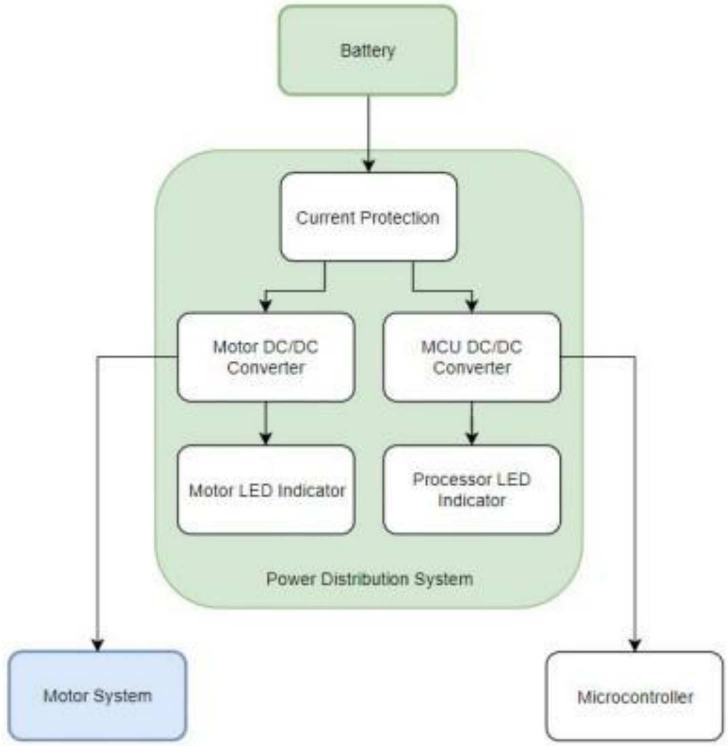
This robot designed to enter a room and seek out a spot where there is extreme heat possibly due to a fire. Upon entering the room, the robot will once again use the color camera to pinpoint a spot where there is a large concentration of light. Once the robot has driven up to the light source, the heat sensor is activated to check and see if there is a large amount of heat being generated. If there is an excessive amount of heat generated, the fan is turned on and rotated quickly with a servo motor to put out the flame. If the flame is not put out the fan will turn on again and continue to blow on the flame. Once the flame is extinguished, the robot leaves the home.



BLOCK DIAGRAM

Power Distribution

As shown below, the robot's power distribution system involves regulating the voltage supplied by the battery to the motor system and microcontroller. Since both systems require different operating voltages, DC to DC converters will be utilized. Although unnecessary on the field, LED indicators will be used for troubleshooting purposes.



Types of fire detectors

Heat detectors • Smoke detectors • Flame detectorsHeat Detectors

- These are generally less sensitive than smoke detectors and are unlikely torespond for smoldering fires.
- They are not suitable for the protection of places where small fires cancause huge losses.e.g. Computer Rooms
- These are suitable for use in places where sufficient heat is likely to be generated and damage caused the heat generated by fire contributes top main hazards. E.g. Battery Rooms, Boilers etc

Water Spray SystemsWATER SUPPLIES:

- Water for the spray system shall be stored in any easily accessible surfaceor under ground lined reservoir or above ground tanks of steel, concrete, or masonry.
- Reservoirs/tank shall be in two independent but interconnected compartments with a common sump for suction to facilitate cleaning and repairs.
- Water for the systems shall be free of particles, suspended matters,etc. and as far as possible, filtered water shall be used for the systems.
- Level indicator shall be provided for measuring the quantity of water stored anytime.
- Water reservoir/tank shall be cleaned at least once in two years or more frequently if necessary to prevent contamination and sedimentation.
- It is advisable to provide adequate inflow into the reservoir/tank so that the protection can be reestablished within a short period.

Pumps

- Pumps shall be exclusively used for the fire fighting purposes. The pumpsused for the fire protection system are of the following types
- Electric motor driven centrifugal pumps, or Compression ignition enginedriven centrifugal pumps or

- Vertical turbine submersible pumps.

– In all the above cases

Fire detection

Fire detection Detection of fire at incipient stage plays very important roleas it enables in suppressing the fire by means of the fire fighting equipments and prevent it from developing in to a major fire.
Detection of fire - visual (presence of personnel is required to communicate to the concerned authorities)
-Automatic (with the use of detectors) Fire Detection system

– This system will provide alarm signal at the initial stage of fire.

– Detectors are located at strategic positions in the area covered by thissystem.
– Detectors are arranged in zones so that the area of fire can be easilyidentified.
– If any of the detectors in a zone is actuated an audio cum visual signal willbe given to the control panel

Necessity of Fire Fighting system

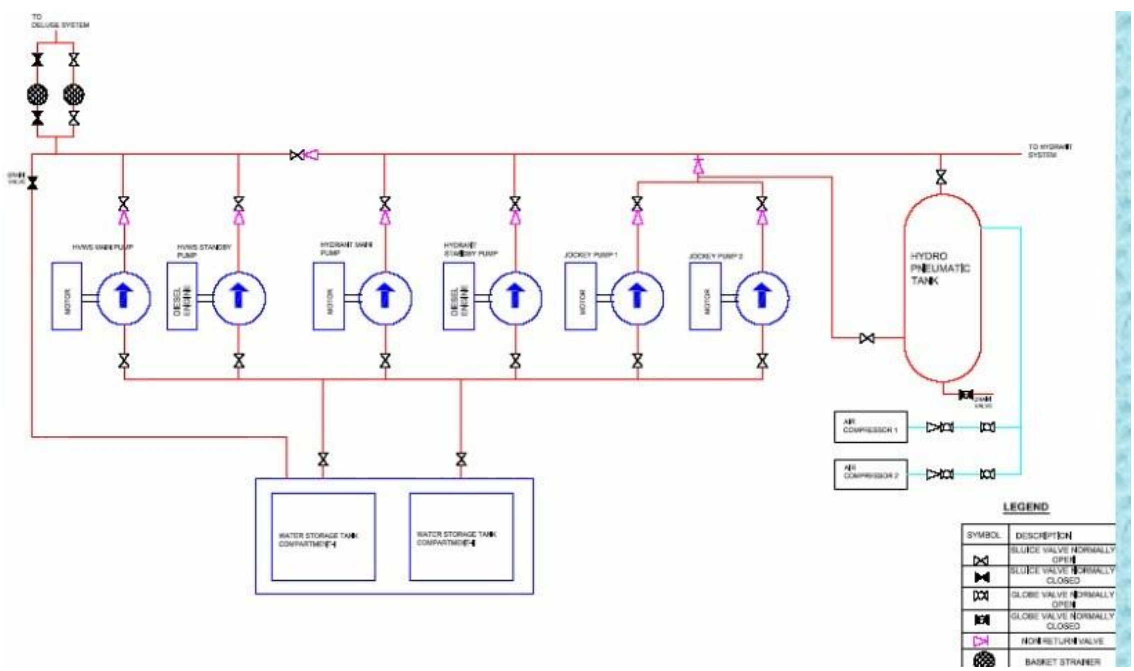
- The fire fighting system in the sub station is very essential

- Saves the equipment from damage

- Loss of life & loss of equipment can be prevented

- Regular trial operation of the system is necessary to detect any fault /deficiency in the system.

Schematic of Fire Fighting



Pumps & Motors • Electrical driven HVWS Pump – 410Cum/Hr Motor125KW • Diesel engine driven HVWS pump - 410Cum/Hr Engine 166BHP

- Electrical driven Hydrant Pump – 96Cum/Hr Motor 30KW • Diesel engine driven HVWS pump - 96Cum/Hr Engine 57BHP • Jockey pump – 10.8CuM/Hr Motor 7.5KW • Air Compressor – 8Kg/Cm2 Motor 3KW

RESULT AND CONCLUSIONS

Result

Firefighting robot (QRob) has been developed to find the location of fire and extinguish it. QRob has an ability to find the location by using flame sensor and ultrasonic sensor. The flame sensor is functioning to sense the location of fire while ultrasonic sensor is functioning to detect the presence of object around the QRob. Both sensors are connected to Arduino Uno, which controlled the movement of DC motor. When flame sensor found the fire, the DC motor will stop at 40 cm from the fire. The operator will be extinguishing the fire using remote control from the distance. The operator also can monitor the QRob by using camera that connects to a smartphone. A. Time to Extinguish the Fire Depends on Distance of Qrob with Fire Source QRob successfully find fire location automatically and extinguish it by operator control. The operator can monitor the location of fire by camera that is connected to the smartphone. Fig.8 shows the time to extinguish fire depends on distance between QRob and fire, and Fig. 9 shows the image during the fire extinguishing process.

Conclusions

Overall, a fire-fighting robot that can be controlled from some distance has been successfully developed. It has advantageous features such as ability to detect location of fire automatically besides having a compact body and lightweight structure. QRob also has the ability to avoid hitting any obstacle or surrounding objects due to its provision of an ultrasonic sensor. The QRob robot can be used at a place that has a small entrance or in small spaces because it has a compact structure. The operator is able to extinguish fire using remote control fromlonger distance. Operators can also monitor the environmental conditions during the process of firefighting by using the camera that is connected to the smartphone. From the experimental results, the robot can sense smokes and fire accurately in a short time. As a conclusion, the project entitled "Development of Fire Fighting Robot (QRob)" has achieved its aim and objective successfully.

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